

Centering Care and Cooperation in the Design of AI Systems with Home Care Worker Cooperatives in the United States

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AI technologies are being introduced to care settings in a top-down way that leads to errors and issues, adds workload, and exacerbates the care shortage. Home care worker cooperatives center democratic governance principles and emerge as an opportunity to highlight workers' voices in technology design. This paper presents results from design workshops with 15 caregivers, administrators, and developers affiliated with a national network of home care worker cooperatives. Participants discussed the tradeoffs between the strengths of AI (e.g., increasing work efficiency, gathering information, communicating across silos, facilitating participation) and its risks and potential harms (e.g., exacerbating caregiver burden, high initial cost, endangering client well-being, adverse environmental impacts). These findings give examples of how to design for human-AI collaboration, diverse stakeholder perspectives, and external relationships or impacts in this context and propose recommendations for prioritizing care and cooperation in other workplaces and organizations.

CCS Concepts: • **Human-centered computing** → **Empirical studies in HCI**.

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1 Introduction

The use of artificial intelligence (AI) is becoming increasingly prevalent, but the people most affected by AI systems often have limited control over how they are designed and governed. When it comes to social computing, AI is often framed as a way to expand access and increase efficiency [15]. However, prior work has shown that technologies built to ameliorate social problems can fail when they do not match the local realities of the communities they aim to serve [89]. These risks are especially important in human services and care settings, where AI may be introduced as a solution to structural problems while adding new burdens, risks, or forms of invisibility [29].

One of these marginalized sectors is home care, which is largely made up of women, people of color, and immigrants [78]. Home care workers support older adults and people with disabilities with daily living and health-related tasks [70]. In response to the high demand for care and low availability of family and friend caregiving, AI and data-driven systems have been proposed for tasks such as matching caregivers with clients [81], facilitating communication across caregivers [75], improving worker well-being [77], and providing companionship and care support [5]. However, research also shows that home care workers are often excluded from technology design, even though they are the anticipated end-users of these systems [76].

Worker cooperative paradigms present an opportunity to explore a different way of designing AI systems. Worker cooperatives are businesses owned and governed by their workers, typically through shared governance structures such

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as boards and member-wide voting [55]. In home care, this model allows caregivers to own the agency, share profit, and participate in decision-making [27]. Prior work has shown that home care cooperatives can improve caregivers' job and care quality in addition to having higher retention and wages compared to traditional home care agencies [28]. Furthermore, since caregivers in home care cooperatives participate in shaping their workplace, they are better situated to question and influence the design and adoption of new AI tools within this context.

Studies have explored AI in cooperatives and care separately, but few have explored their confluence. Prior research has explored how AI is becoming more integrated into the cooperative movement [33, 44]. For example, it has raised tensions such as the one between the transparency required to maintain accountability to members and the need to protect intellectual property rights from competitive market actors [69, 83]. While home care has similar issues around worker voice and insight into agency operations, workers often do not face the same issues accessing sensitive and private data from clients [61]. These differences stem from the unique power dynamics and relationships in home care, which commonly manifests as an alignment in the relationship between workers and clients and traditional contention one between the workers and their agencies. Thus, this intersection presents a unique opportunity to explore how concerns that stem from cooperation versus care may build on or differ from each other.

This study builds on the extant literature on home care worker cooperatives to examine how stakeholders imagine AI in this setting. We conducted design workshops with caregivers, administrative staff, and cooperative developers in U.S. home care worker cooperatives. During the workshops, stakeholders discussed their current impressions of AI and brainstormed AI implementations to address problems in caregiving and cooperative operations. The purpose was to examine stakeholders' current perceptions of AI and recommendations regarding the design of AI systems for use by home care cooperatives.

We find that participants perceived AI as a potential means of closing human and capital resource gaps and increasing efficiency in tasks such as scheduling and administration. However, they also emphasized that AI itself requires resources, including time, training, infrastructure, and human oversight. Participants also imagined AI as a tool for gathering and analyzing data but cautioned against the negative effects of inaccurate information on safety, privacy, and client well-being. Although participants saw AI as a way to foster communication between caregivers, clients, and cooperatives across the boundaries of distance and isolation, they worried it may weaken human connections or cause environmental harms. Finally, participants identified AI as a possible tool for strengthening worker voice in decision-making but emphasized the need to democratize its ownership to align with the values and principles of cooperativism.

This work collectively contributes to the understanding of domain-specific examples of how caregivers, administrators, and developers imagine AI systems in home care cooperative contexts. Moreover, it begins a nuanced discussion of how it might be possible to balance values of care and cooperation with AI systems. This research is especially relevant for other contexts that are at the intersection of being democratically governed and moving towards collective identity, solidarity, and decision-making as well as have orientations towards promoting well-being and connection in society.

2 Related Work

While AI technologies are poised to make changes towards "social good" (e.g., in global education [6], healthcare [37], civic engagement [46], and environmental justice [79]), there are still challenges matching proposed AI solutions to the sociocultural [89] and local contexts of marginalized communities [64], as well as organizational challenges in adopting [3] and implementing [49] AI solutions. This misalignment in values stems from a long history of the failure of emerging technologies and interventions to adapt adequately to the needs of their intended beneficiaries [85] to

solve “wicked problems” [71] without generating new ones [56]. Moreover, more research has challenged whether technological artifacts are inherently neutral [95] and how best to adapt and account for values like inclusion and participation [25]. AI systems complicate these tensions by introducing externalities beyond the scope of individual- or organizational-level interactions, including entrenchment of labor [94], bias and misinformation in the broader information ecosystem, and environmental concerns [51].

Therefore, it is important to explore these tensions, especially in settings and organizations that face resource constraints. Many organizations feel pressured to implement AI technologies to keep up with the latest trends or because their bosses tell them to. In other cases, data-driven and algorithmic systems are adopted at a higher industry level, such as government mandates for certain electronic work tracking systems that use data-driven models [52]. Thus, in this situation where many organizations are cornered into using AI systems, whether by necessity or top-down directives, it is important to carefully examine how these tensions could be balanced in AI systems that are able to achieve their socially intended outcomes. The relevance of specific tensions depends on the context in which AI is being implemented. In this study, we investigate home care worker cooperatives, which exist at the intersection of two areas that pose challenges for the design and implementation of AI systems: frontline care work and cooperative movements.

2.1 Complications in Accounting for Care in AI

Home care in the United States refers to paid, professional caregiving that supports 7.3 million older adults and people with disabilities living at home, provided by a 3.2 million person workforce, the majority of whom are women, people of color, and immigrants [66]. Home care workers provide assistance for daily living (e.g., bathing/grooming, cooking/light cleaning), social support (e.g., companionship, maintaining engagement in community activities/social networks) and clinical monitoring under remote or intermittent supervision (e.g., checking vital signs, going to doctors appointments). In frontline care, and home care specifically, AI systems are cropping up to address the high demand for at-home services paired with the low supply of family and friend caregivers [76]. The AI tools currently available help to distribute care (e.g., matching caregivers and clients [20]), perform care tasks (e.g., caregiving robots [96]), guide caregiver decision-making (e.g., interactive voice assistants [5]), foster communication across different types of caregivers (e.g., [75]), or substitute for caregivers when necessary (e.g., phone calls to socially isolated individuals [39]). Likewise, data-driven and algorithmic technologies have been developed to support family and friend caregivers (e.g., tools to promote work-life balance [12]) and paid care workers (e.g., improving their health and well-being through sleep/movement activity tracking [77]), as well as to counteract worker exploitation (e.g., tracking wage theft [61]).

However, prior proposed technologies have often been not well matched to home care settings. Even though the home care workforce provides essential care for vulnerable clients, they are often overlooked and devalued, with many facing high levels of emotional [80], physical [78], and financial [58] stress. This invisibilization is further perpetuated by the design of technologies that do not fit into the realities of home care, as the perspectives of home care workers are often not considered [87], which causes additional technostress and technological burden [59]. Examples include what Wright [97] describes as failures of “care algorithms,” where instead of supporting human caregivers, the systems end up causing them to have to spend more time fixing technological issues and less time giving care to their clients. Ming et al. [60] delineate two main ways AI tools are often misaligned for frontline care settings, namely that (1) care itself is difficult to quantify, and that (2) care often implicates broader societal well-being and dynamics across many stakeholders.

Many scholars have explored the potential for considering care in relation to sociotechnical systems, some of whom consider assemblages of human and nonhuman actors [14] or how the values and ethics of care are leveraged [86],

including across HCI literature [65]. Other scholars have examined how these broader principles of care apply in caregiving settings [38, 40]. Our research expands on conversations about incorporating care into technology design for caregiving settings, specific to home care.

2.2 Aligning Cooperative Values with AI Design

The International Cooperative Alliance (ICA) defines cooperatives (or co-ops) as “*an autonomous association of persons united voluntarily to meet their common economic, social, and cultural needs and aspirations through a jointly-owned and democratically-controlled enterprise*” [36]. This paper deals specifically with worker cooperatives, which are “*business[es] owned and controlled by [their] workers,*” where worker-owners both contribute their labor to the cooperative and have representation or voting rights (i.e., “*one worker, one vote*”) for the governing body (e.g., board of directors) [88]. There are worker cooperatives in many sectors, including retail and service, manufacturing and skilled trades, technology, and home care (e.g., you can find a worker cooperative local to you if you’re based in the United States here: <https://www.usworker.coop/directory/>).

AI is starting to become more integrated into the cooperative movement. For example, Siebert [74] calls cooperatives to “*embrace the opportunities of AI,*” arguing that AI tools would not only make cooperatives more effective, but also introduce cooperative governance models into AI. Some examples of integrating paradigms like public interest infrastructure (e.g., computer and storage power owned and operated by nonprofits, government, and universities), data cooperatives (e.g., individuals having control over who can access their data, what data they can access, and for what purpose), and collective governance (e.g., through civil society, citizens assemblies [34]) include AI cooperatives or platform cooperatives (e.g., READ-COOP that develops and hosts its own AI and ML Automated Text Recognition [83]).

Some HCI literature has critically examined cooperatives as a site of technology development and deployment. There were challenges around aligning the cooperatives’ values of transparency when default systems designs often protect or hide financial details [93]. Additionally, resource constraints resulted in competing demands for members’ time and a “*community artifact ecology*” encapsulated a piecemeal amalgamation of technology [43]. Finally, cooperatives may have difficulty properly representing themselves both to their members (i.e., social and personal benefits of participation) or the public [19]. These difficulties draw from a clash between the values of the cooperatives and the technological tools to which they have access, especially given their unique governance compared to other social good organizations.

There is also HCI literature that explores how the values of cooperatives can be increasingly incorporated into design. Verma et al. [92] summarize these efforts as integrating the two intertwined values of “*cooperation and commoning,*” including pointing to “*commons design*” as a way of emphasizing relationality and social justice. They even discuss how the origins of participatory design were rooted in worker justice and giving workers voice over their workplace technologies as well as their working conditions [26].

2.3 An Empirical Investigation of AI in Care and Cooperation

Our study contributes empirical evidence about a real world context where frontline care and cooperatives intersect, building on prior research regarding the unique intersection between caring and commoning as a way to confront platform capitalism [73] and the challenges in caring for the commons [23]. Both care and cooperation require strong relational ties in order to provide meaningful support or connection [59] and develop consensus on organizational questions [93]. Moreover, they both have broader aims beyond individual interactions to impact the broader care ecosystem [60] and address economically and sociotechnically exploitative systems [92]. Our study allows us to not

ID	Role/Position	Gender	Race	Education Level	Age	AI Experience
AC1	Admin/Caregiver	Woman	White	Some college	30-39	Regular, non-expert
AC2	Admin/Caregiver	Woman	White	Some college	40-49	Regular, non-expert
AC3	Admin/Caregiver	Woman	Black or African American	Associate's	50-59	Via existing apps
AC4	Admin/Caregiver	Woman	White	Associate's	50-59	Tried a few times
C1	Caregiver	Man	White	High school/GED	60-69	Via existing apps
C2	Caregiver	Woman	White	High school/GED	50-59	Regular, non-expert
C3	Caregiver	Man	White	Bachelor's	60-69	Regular, non-expert
C4	Caregiver	Man	White	Graduate/professional	30-39	Via existing apps
C5	Caregiver	Woman	White	High school/GED	20-29	Via existing apps
C6	Caregiver	Woman	Black or African American	Some college	70-79	None
D1	Developer	Gender fluid	White	Graduate/professional	40-49	Tried a few times
D2	Developer	Woman	White	Bachelor's	40-49	Regular, non-expert
D3	Developer	Woman	White	Bachelor's	50-59	Tried a few times
D4	Developer	Woman	White	Graduate/professional	50-59	Regular, non-expert
D5	Developer	Woman	White	Bachelors	30-39	Via existing apps

Table 1. Self-reported demographics for each participant, including job role/position, gender, race, highest level of education, age, and AI experience.

only tease out the overlap between these two concepts, but also grapple with what it takes to include both of them and the challenges that arise in reconciling them, guiding future AI design to infuse the values of care and cooperation.

3 Methods

3.1 Participants and Recruitment

We conducted an IRB-approved study from March to August 2026 with stakeholders from home care worker cooperatives across the United States. We recruited a total of 15 participants: 6 caregivers (C) who were involved in directly giving care to clients, 4 acting as both administrators and caregivers (AC) that also took on office or operational tasks in their cooperatives, and 5 cooperative developers (D) whose jobs are to help people start, expand, or convert democratic, member-owned cooperatives and focused specifically on home care cooperatives. These participants were identified using targeted recruiting and snowball sampling through a national network of home care worker cooperatives, some of whom were follow-up participants from an interview study about mechanisms for participation. Overall, the participants were mostly women or gender fluid (80%) and white (86.7%). Most had more than a high school education (80%), with some having graduate or professional education (20%). Additionally, there was a range of ages, with a little less than half (46.7%) under 50 years old and a little more than half (53.3%) 50 years and older. They self-reported a range of AI expertise, with most using AI only through an existing application (33.3%) or having little to no AI experience (26.7%). The rest reported using AI regularly but did not consider themselves an expert (40%). For more details about the demographics of the participants, see Table 1.

3.2 Study Procedure

Participants first completed a survey, which included screening (i.e., if they were over 18 and involved in home care cooperatives), informed consent, and the collection of demographic information and knowledge of and experience with AI. They then participated in design workshops that lasted approximately 120-150 minutes. These design workshops were modeled after futures workshops [22], speculative design [40], and participatory design sprint methodology [16]. This approach was selected to first understand participants' preliminary perspectives of the broader promises and pitfalls of AI, setting a foundation from which all conversations could arise. Each workshop included one to three participants, with the goal of grouping participants with similar positions and AI experiences.

The workshops were divided into two parts. In the first part, participants discussed their experiences with and impressions of AI. This was followed by a presentation introducing a basic definition of AI and examples of the use of AI across different contexts to ensure all participants approached the conversation from a shared understanding. This portion of the workshop was intended to show participants that although AI has received significant attention in recent news, many of the technologies underlying AI have existed for some time. The presentation also included AI's capabilities, limitations, and biases to set up the later design activity of imagining how AI could be used in home care cooperatives. This discussion also touched on the potential harms of AI at an individual, group, and organization level, as well as ways to respond to these harms through collective bargaining and participatory design, which aligned with the broader goals of the study. During the presentation, participants were asked to respond to interactive prompts to help them relate the presentation to real-world experiences and the home care worker cooperative context.

The second part of the workshop focused on the design process, using participants' expertise in their work to explore how AI could potentially fit into the home care worker cooperative context. The workshop focused on four of six major steps of a design sprint process [1]: understanding the issues, defining a problem to be solved, sketching possible solutions, and deciding an approach. Participants began by discussing different issues they face related to cooperatives and caregiving work. After narrowing down critical issues for further discussion, participants were instructed to generate ideas for addressing the identified issues using a "magic wand," inviting them to imagine potential worlds in which these issues were addressed and paths they might take to actualize these possible worlds if there were no limitations of resources or technological capabilities. Speculative futuring practice such as these are common in design workshops [62]. Participants were invited to express their ideas through words, images, lists, sketches, or stories, allowing them to explore different possibilities for integrating AI into the current home care context.

At the end of the workshop, participants completed a follow-up survey to understand how their perspectives had changed during the workshop as well as their key takeaways for the future use of AI in home care.

3.3 Data Analysis

Workshops were transcribed using Zoom, corrected against the audio recordings, and de-identified by one of the researchers. Each of the workshops was then analyzed using thematic analysis following guidelines from Braun and Clarke [7], with emphasis on identifying and reporting patterns within data. Three researchers independently reviewed and coded the transcripts and then met to compare interpretations, discuss differences, and refine a shared set of 55 codes (e.g., "*comparing AI and human*," "*talking about small business problems*," "*using AI for safety*") that were developed using gerunds, following Charmaz [11]. We then used an inductive approach to allow themes to emerge from participants' discussions while also attending to concepts related to AI use and home care work, with related codes clustered into potential themes and representative excerpts selected to illustrate each theme.

3.4 Positionality and Methods Reflection

Our team is composed of a combination of researchers with expertise from a variety of disciplines, including human-computer interaction, computational social science, human factors, social welfare, and health services research. Three of the authors have worked closely with home care workers and/or home care worker cooperatives in the past. Some have served as subject matter experts who have co-designed or co-developed AI programs (2 of 7) or developed AI programs themselves (1 of 7). The remaining have used AI regularly but do not consider themselves experts (3 of 7). The majority identify as not cisgender men (4 of 7) and as not white (4 of 7).

4 Findings

Participants were excited to discuss how AI could help them in their home care worker cooperative spaces, but they also raised some concerns. As caregiver C1 put it, AI was seen as a “*double-edged sword*” that had both potential benefits and harms. However, many were interested in discussing how tradeoffs are made on a broader scale but also individually, “*so that it can be a calculated decision—for any task—do I use AI for this?*” (D2)

4.1 Balancing Technology and People Resources in Operations

As the cooperatives we spoke to were small businesses themselves, many saw AI tools as a way of helping them close resource gaps by enabling them to do their work more efficiently. For example, some participants spoke about the ability for AI to help get information (e.g., “*comparing how many caregivers do I have versus my competitor, what am I charging versus what they’re charging, what tasks do they do*” (AC2)), translating information (e.g., turning local administrative code legal jargon into “*human language for me, which is wonderful*” (AC2)), generating visuals (e.g., “*creat[ing] a picture in my space that I provided and added artificial people and images to it...[as] one of the definite positives for marketing*” (AC1)). For communications, others used AI to express themselves and find the “*proper wording*” (AC2) in English or to “*keep thoughts on track and convey them in a focused way*” (AC1). In sum, administrator AC4 described,

“In our area, [AI]’s got a really bad reputation. I just think there’s more sides to it than the bad. In our business, there’s things [for which] it has been very beneficial and helpful. So that as a small business, I don’t have to hire someone else. I can go in and do ChatGPT or whatever and [it can] help me with some of the stuff I need to get done.” (AC4)

Saving resources as much as possible was especially relevant in home care settings because, as D1 described, “*one distinction between other co-ops and home care co-ops is just the very small margin that home care co-ops are operating in and how difficult that is.*” They went on to describe how this was also compounded with federal investigations and changes to Medicaid funding. Additionally, being cautious about resource gaps was especially important due to the short supply of caregivers and the need to juggle resources between caregivers and clients. C1 referred to the intertwined problems as a “*the chicken and the egg.*” AC4 called it a “*constant back and forth.*” And AC2 wondered, “*how do I get qualified caregivers so I can get more clients to get more money to fulfill the needs of everybody?*” Developer D5 described how this cause-and-effect dilemma strained organizational resources as well, making it difficult to maintain the creation of cooperative ideals, and proposed AI as potential stopgap for resources:

“[(AI)] creates significantly more efficiency. That’s one of the greatest challenges that I see for home care co-ops is the more volume they’re taking on, the less time they have to develop democratically... So reducing operational oversight allows for a continuation of maintaining that other side of things.” (D5)

The care workers themselves were already doing many different jobs, as C6 listed, “*you gotta do the laundry, you gotta run and pick up the meds, you gotta go to the supermarket and get stuff, and then ... you should stop with the cleaners.*” Furthermore, C6 discussed how the demands she faces as a caregiver meant she had less time to spend on other tasks, such as learning new technologies. She said that, with regards to AI, she’s “*never really used [AI], because most of the time is spent on [her] job.*” She was mixed on new AI implementations because on one hand, she saw it meant “*that would be a whole other training for us where they would have to take us off a job*” but did also think it was a good idea it would ultimately improve the efficiency of giving care to clients and “*help them as much as you can faster.*”

As many of the caregivers and workers were themselves older adults, including a developer who said that she herself was “*terrified*” (D4) when she thought of AI or another who discussed how some of the caregivers they work with are

“very anti-technology” and “old school...like tracked the schedules of 20-plus caregivers in a notebook” (D2). C2 reported that her cooperative even had “two caregivers that don’t even have smartphones.” C1 felt that in implementing AI solutions, “it’s beyond my skill set, and I don’t have the time or energy to acquire those skills.” Ultimately, administrator AC2 saw handling this emerging technology as something that could be handed off to the next generation—a transition for which AI could prove helpful:

“My older caregivers are burnt out. They’re tired. They are just tired of carrying that torch as long as they have. And they’re continuing to, but I can feel hesitancy on picking up more things. I can feel resentment when I ask them to please do this thing one more time. they’re tired, so getting that younger generation to understand how important it is... So if AI can fix that, I would be all on board.” (AC2)

One specific area many participants focused on in terms of quality and efficiency of administrative functions was scheduling shifts, which involves matching caregivers with clients. For example, caregiver C5 noted the difficulty of the time- and resource-intensive undertaking that involves “over 100 clients, over 100 employees” and admins who are “laying out schedules across a big conference table, working to the last second.” In the past, this process led to high uncertainty (e.g., “I’m waiting until the day before...to figure out my...schedule”) or mistakes (e.g., “I’ve driven 30 minutes to a client’s to show up and they’re not there because they didn’t realize that they had care scheduled that day because the schedule changed and nobody told them.”). She suggested AI as a tool that could help “take a load off that massive scheduling job” and have “optimization and efficiency [so] caregivers and clients can both have their schedules more than a day or two prior.” However, she also noted that it was important to “**always, always, always, have it verified by the humans.**” This highlights the tension between the efficiency gains of new technology and the need for a human touch to ensure that implicit knowledge about preferences and personalities are incorporated in the assignments.

Therefore, home care cooperatives are presented with a conundrum: AI could potentially help them close resource gaps but requires digital and physical infrastructure and resources in terms of people, money, skills, and training. Home care cooperatives often enjoy advantages in juggling resources with what D1 calls the “co-op difference,” or having better working conditions and, therefore, less turnover and higher quality care because workers have input into the governance of the organization. Many caregivers prized their connection to their job, talking about how “this is very human-centered work” (C4) and appreciating how their cooperatives treated them “we aren’t just a number for them” (C5). Thus, with new AI systems, it is important for them to also consider how they are balancing their technology and people resources.

4.2 Ensuring Safe and Private Information in Caregiving

One way AI was described as being useful was in the gathering and analyzing of information related to caregiving. Many participants shared examples of how they used ChatGPT and other genAI tools to gather information on how to perform basic tasks (e.g., using a different type of catheter (C6)) or respond to specific cases (e.g., statistics about warning signs in diabetics (C3)).

Moreover, participants said AI could be used as a mediator between stakeholders involved in the home care ecosystem. C6 described difficulties with medical professionals who “question you because you’re there, not the family member” and the family members who get confused due to their absence at appointments where the doctor changed a medication regimen. She saw a role for an AI tool that could help mediate communication across the different caregivers for accountability and alignment. C1 pointed out the importance of patient advocacy especially for people who have to contend with “diminished capacity, plus all these terms and technology, and [also] having to try to get through the medical

centers.” He proposed a tool that could share “*a photograph of her condition in real time as well as a transcript of what the doctor said so that everybody can be on the same page literally.*” AC3 mentioned that that AI could play a role in emergencies in contacting with loved ones, qualifying that suggestion by saying “*I don’t know if AI is necessarily an answer, but I know that it could definitely help.*”

However, one big area of concern was the correctness of the information gathered or shared by AI. Many participants shared this concern and highlighted that wrong information “*can be really dangerous when you look at caring for others*” (AC3). C5 expanded on this further:

“There could be confusion with AI getting something wrong medically, with a diagnosis, with a medication, with a dosage... That kind of scares me... We have to be picky because there could be a lot of benefit to having AI used as some of the more hands-off stuff, but we have to be careful how far in we’re letting AI in, to protect everybody involved, because AI is wrong a lot.” (C5)

She also pointed out the potential issues with legal limitations and suggested that AI not be used “*where it could be harmful*” like in circumstances that require direct patient interactions. Additionally, C6 explained that a lot of what goes on in care is a matter of “*life and death*” and happens really quickly. She was also concerned about the speed at which AI would be able to assist in an emergency situation because when something did happen on the job, she has had to “*make a spot decision*” and did not have time to search for information or even contact her agency. C3 also raised the concern that “*if people lean too heavily into AI and there’s a mistake made and there’s nobody there to monitor it, that could be detrimental to a client.*”

C3 was also concerned about the potential for AI-driven approaches to result in an “*infringement on [the clients’] privacy.*” This highlighted another important concern about AI information gathering and analysis—that in addition to the implications on the safety and well-being of the patient, there were also issues around confidentiality. Many participants, including C4, noted the importance of considering whether the AI tools were “*HIPAA compliant,*” or adhering to federal regulations about the sharing of sensitive patient health information. Others like D2 questioned whether publicly available AI tools had her personal or organizational best interests in mind, wondering “*what protections do we actually have?*” Admin and caregiver AC4 also noted concerns about clients who, “*instead of going to the doctor like they should...they’ll just look it up on ChatGPT.*” She found this “*extremely dangerous*” given risks associated with the security of client information shared with big corporations and with the questionable accuracy of information accessed by clients.

Thus, the participants pointed out how helpful AI tools could be for getting and sharing information, but also shared challenges in running up against the “*artificial limits to information*” (C4). C5 said that she has experienced these limits when interacting with clients in her other jobs who “*take the things that AI says and hold them as truth and nothing but the truth.*” There were also concerns about AI becoming “*too realistic*” (C2) and whether it would take advantage of older adults who are especially “*vulnerable*” (C3).

Some of the participants shared a few of their practices for combating misinformation, which included using AI “*as a first pass for getting references that you can then check out yourself and evaluate on your own*” (C4). Others shared concerns about the tension between more specific prompts “*so it doesn’t go off on ‘society norm’*” with all its biases (AC2) and generalized prompts that go beyond the limitations of the “*tailored answer based on the information you’ve given it*” (AC1). Furthermore, two participants discussed concerns that the “*algorithms might intentionally be biased*” (GL23). C6 noted the importance of making sure AI tools were safe, suggesting that, for any proposed AI implementations, it would be best to “*test run it first.*”

4.3 Pushing the Limits of Virtual and Physical Connections

AI was seen as being able to bridge gaps in the physical world through virtual connections. Because caregivers are often traveling to client homes to provide care, they do not always have physical workplaces to gather and interact. D2 explained that an attribute **“unique to the home care space that some [other types of] co-ops don’t have is the isolation.”** At the same time, she also described how cooperatives had a propensity for bringing people together towards shared values, which meant that **“co-ops naturally...combat the isolation.”** Some of the participants were interested in seeing how AI tools could bring the workers closer to each other feeling that **“once we get the tools, we can help each other (C6).”**

The cooperatives themselves felt isolated from each other and the outside world, with many noting they were based in really **“rural”** (AC4, C3) or otherwise remote areas. For this reason, they really prized opportunities to connect. Some of the participants had actually met before because they were in similar geographic areas starting similar ventures. Others of the participants even noted how they enjoyed connecting with other caregivers or admins during the workshops. For example, AC1 noted she **“really enjoy[ed] this whole thing and being able to talk with other like-minded people that are having the same difficulties and same age-issues with technology.”** Others also enjoyed the **“wide variety of perspectives”** (C4). The participants wanted to leverage AI tools as a way to connect with others in their similar situations and share their experiences. C1 said:

“I wish I had something to share with you that you might be able to make available to other home care co-ops. And maybe some of them are doing something that you could share with me.” (C1)

Thus, the participants were excited to see connecting with other cooperatives as a potential use case for AI given their need to interact more with other cooperatives and their positive experiences forming such connections in person.

In addition to helping cooperatives connect to across physical isolation and combat information silos, AI tools were also seen to have promise in connecting isolated clients to each other. C1 has dedicated his career to **“providing some semblance of communication and companionship...[so that] people aren’t isolated in their homes,”** promoting different ways of gathering people in conversation about difficult topics like death and dying. He saw AI as being able to help clients toward **“companionship or social interaction”** (C3) and to be seen or heard by **“other friends that are supportive”** (AC3). Others mentioned emerging AI-powered applications aimed at improving companionship, like conversational agents, and discussing how that might be better than humans in some cases. C1 spoke about one benefit of AI, where **“AI wouldn’t lose patience with those folks like a human might”** and AC2 said **“AI would be calm, collected, and whatever tone you told it to be every single time.”**

However, there were concerns about AI-facilitated or AI-only companionship. As AC1 said, **“I still fear for the personal connections that humans may make with AI and lose some humanity and some of that socialness that we as creatures need”** C3 discussed how he personally preferred interacting with administrators and staff at his cooperative in person because **“...I don’t like talking about stuff like that on the phone. I’d rather walk in and talk to someone personally because it’s easier to explain and then I can use my hands.”** D2 raised a concern, saying **“I don’t think AI can or ever will be able to do what a human can do in terms of relationship building.”** AC2 echoed D2’s concern, noting the importance of maintaining the human side of care:

“In the home care world, I believe human connection, compassion, and personal care should always remain the main priority... AI should be used as a tool to support people, not replace human judgment or relationships.” (AC2)

There were some cases that were brought up where AI would not necessarily replace humans, but fill in care gaps between human caregivers. One example C1 gave was to have a device that could “*take readings on the client when we’re not physically with them*” which would then “*give us some clues that we could use*”. AC2 mentioned also that these tools could help improve the well-being of clients who were extremely in need:

“Because we are human, as much as we would love to cover 24-hour shifts, it’s not always feasible. So having that little bit of hey, let’s do 3 hours with AI watching them until we can get the other caregiver in there is a better alternative than nothing” (AC2)

C3 gave an example of something that could also interact with the clients’ physical environment like their climate controls or phone in a “*voice activated device if they need help*.” Another example C3 gave was using AI to help identify issues in physical spaces and propose potential solutions. C3 said as a caregiver he was very invested in the safety of his patients and making sure their homes had clear pathways without tripping hazards like throw rugs. He noted some specific examples of how AI tools might be helpful for helping clients re-organize their spaces to be safer or help clients find the “*safest way possible for them to navigate through there*.”

While AI was seen as being able to bridge some physical gaps, potentially in the absence of caregivers, there was also debate on the limits for AI to be able to have a presence in the physical world. C6 pointed out the limitations of AI in performing more complicated tasks, saying, “*I don’t know if the robot would be able to get that catheter onto a person...you still need a human being*.” C5 and C3 also spoke about the difficulty of AI when performing “*hands-on labor*,” especially for the provision of care directly to the client. As developer D2 pointed out, it is important to have physical interaction from human workers in order to bring plans to fruition, noting that “*AI can be a thought partner...but it certainly isn’t a replacement for good old-fashioned elbow grease*.”

Another element that the physical environment brings into interactions with others and with AI tools is unpredictability. One example that was given was the transportation issues caregivers faced since “*most of us, we don’t have our own transportation*” (C6). The uncertainty of public transit schedules and traffic conditions meant that figuring out which workers would be when and where would end up being a “*logistical nightmare*” (C5). The participants noted that AI systems that would be able to take these interactions between the real-world and the virtual schedules into account would be beneficial. Specifically, “*being able to plug in...everybody’s ever-evolving schedule and availability...[with] the ability to flip and to pivot and to change*,” would make using AI “*a lot easier than a traditional algorithm*” (C5).

In addition to discussing the impacts of the AI virtual world on their own, real-world, physical spaces, the participants were also acutely aware of the impacts it could have on the broader environment and infrastructure. C5 was one of the first to bring up the environmental costs, stating:

“It doesn’t matter if we’re using it for good or bad, it’s still impacting our environment in a huge, huge way that we have to be cognizant of... No matter what the benefit is, we have to be aware of what it’s doing to the land and the air around us... I think it could work if it’s done right, ethically, morally, and environmentally friendly.” (C5)

Specifically, she pointed out the impacts that she was seeing in her own local physical environments, saying, “*I see it in my own community, I see it in my own state, and it’s terrifying*.” Moreover, D1 was talking about their own positionality “*living in the desert where the river has been dry for the past two months*” and their wondering “*can this technology be environmentally sound?*” In addition to their local environments, others mentioned the impact on local communities in their areas such as “*mass surveillance and AI-supported technology that our police department is using*” (D3). D2 said it’s

important to consider “*what’s the impact on the environment, on the communities we care about more broadly?*” and that she was “*worried that...the end doesn’t justify the means.*”

Therefore, even though AI technologies had the opportunity to help the participants cross physical and virtual boundaries, they were also concerned about ways the physical environment impacted and was impacted by technologies. Ultimately, the participants wanted to understand the cost-benefit ratio, suggesting that it would be helpful to have “*clarity on what the actual impact is*” (D2) and the ability to calculate the tradeoff so that she could pinpoint “*the things I should be worrying about and need to worry about more.*”

4.4 Raising Worker Voice and Challenging Structural Power

Finally, the participants discussed how AI would be able to help build and raise worker voice. One form of worker voice was within the cooperative itself, which included employee rights and grievance voice as well as collective or representative voice. Many of the participants described how home care worker cooperatives work, including how decisions are made, with most made by the executive board (i.e., a voluntary group of administrators and caregivers that meet every month or so) and some brought to a vote by the whole membership of worker owners. C6 also mentioned her involvement with both the board and the union as a way of “*representing my coworkers on both sides*” and how that involvement was how she became aware that she was being underpaid.

Both C6 and C2 spoke about different examples of ways the cooperatives created channels for worker voice (e.g., weekly union desk at the agency office, open board meetings any member could attend), but both felt “*we definitely need more people involved*” (C2). Some of the barriers to participation identified included workers feeling unsure or insecure about their knowledge or language skills as well as others being worried about getting in trouble given their citizenship status. C2 said that she thinks speaking up in the workplace “*scares a lot of people, but it shouldn’t.*” She noted the importance of having “*the courage to ask questions...even if you don’t know how to phrase it correctly, you still need to ask, because someone else in that room is thinking the same thing and they don’t want to speak up either.*”

Another barrier was interpersonal dynamics. AC3 described an incident where she felt she was being misrepresented by a co-worker, and said “*it still bothers me to this day because a lot of that could have been avoided had somebody asked me a question instead of going behind my back*” (AC3). This led to her being more cautious about how she interacted with her co-workers and her concerns about being “*way off in left field or being uncooperative.*” She also mentioned using LLM tools to help her process the interpersonal conflict, typing in the conversation “*verbatim into the computer.*” She found it helpful to “*have a captive audience who isn’t just gonna tell me what I want to hear, but is gonna tell me a number of different ways that it can be looked at.*”

One of the most discussed barriers was caregivers understanding how to be worker owners. D1 said “*that’s a really steep learning curve, to be able to understand profit and loss statements and cash flow,*” especially because, as D4 put it, “*their experience is in the caregiving role rather than the business management role.*” Given that caregivers were also charged with administrative and operational tasks as worker owners, they often had to balance these organizational concerns with their caregiving. This led to tensions because, as AC1 put it, “*Emotions are great. I love my clients. I cry for them, I feel with them. But that feeling does not pay my bills.*” Moreover, C2 spoke about how the focus on the “*bottom dollar*” was difficult for caregivers who had to juggle both care and business and wear different “*hats*” in different situations:

“*We are in the business of taking care of people. But when you are a member owner, you need to sometimes take that hat off as being a caregiver... when you’re in the meeting, you are an owner of the*

business. So you need to do what's in the best interest of the business. It might not always make every caregiver happy, it might not always make every client happy" (C2).

C2 recounted how, overall, "it's a difficult subject to teach somebody what it means to be a member owner" and feels that "AI could help as far as that goes."

Others had ideas for what an AI-driven tool that supports training for member owners could look like. C1 saw it as "a chatbot that had a home care cooperative-specific knowledge base." AC3 suggested "a flash drive that a company or co-op could purchase and it would come with the basics, which would be that extensive knowledge base of...rules and regulations, a broad base of home care information, and training courses, and so forth." D4 gave the example of adapting "the home care launch box" already being developed to different geographic considerations, a concept which D3 referred to as "an interactive roadmap." D2 suggested different tools for helping with "financial literacy" and D3 suggested having "an AI tool that would help with the modeling and projections and decision-making points." However, the participants also noted that they did not want such a tool to take away from the relationships in the "cooperative ecosystem" (D3). D5 described an example of how she leveraged her own network when she was just starting out. Working with the experts "wasn't just an educational process, it was relationship building."

Even with the opportunity to use AI in the cooperative structure to improve participation of caregivers, there were still concerns. For example, C6 was frustrated with the cooperative structure in her cooperative, which was larger and more developed than the other cooperatives mentioned by participants:

"We are just rubber stamps within this organization, because you all done decide whatever it is, you do it, then call the meeting and say, oh, ladies, we come up with this idea... So we bring it to you guys now for you all to give us the okay. But I would say... you were supposed to come to us first." (C6)

Many were frustrated that even though worker cooperatives were a way to circumvent the strict power hierarchies, there were still barriers to participation and residual challenges. Similarly, they were concerned about what AI would be able to do to change these dynamics. As C6 asked: "What are the benefits? **Would it help us? Or just you guys?**"

These entrenched power structures extended beyond the cooperatives. Many lamented the ownership and power structures around AI and how those decisions were being made and the benefits distributed. Some like AC3 were concerned about the "people who are poor, people who are elderly" because "if they fall through the cracks, they're gonna get eaten up by this society, and it's not something I want to ever see happen." She suggested that AI might be able to help but would not be the full solution for this power imbalance. Participants like C1 discussed some of his concerns for how AI was currently owned and governed and wanted to explore alternative forms of governance:

"My view is that AI is using the hell out of us, so we should probably figure out how we can get at least some benefits, because these people wielding all this power are out of our grasp. So what can we do with what's in front of us in our community? ... It shouldn't be surprising that a home care cooperative would want a democratic form of administration for the use of AI" (C1)

He further went on to explain his hope that there would be mechanisms for distributing some of the benefits of AI systems more broadly across society, such as a "technology dividend paid to every citizen, a portion of this incredible wealth that's being accumulated by this technology" (C1). Many of the other participants also called for more "democratic accountability" (C1) for how AI is being leveraged and who gets to benefit.

Potential Benefits	Potential Harms	Opportunities
4.1 Balancing Technology and People Resources in Operations		
- AI can support administrative functions (e.g., getting/analyzing data) that are otherwise very expensive for small businesses (i.e., requires hiring more people) - Especially helpful in home care, a field that has small margins and a shortage of workers	- Current workers are already overloaded with responsibilities - This increases the cost to learn new technologies, especially for those who are less immersed in technology (i.e., less tech-savvy) - And increases the cost for maintaining (necessary) human involvement (e.g., operating AI, checking AI's work)	- Making it easier to calculate the inherent costs for changing to a new system (e.g., for training the workers, for worker labor needed to monitor the systems) - Developing design, deployment, and maintenance strategies that reduce these technology and people costs (e.g., employee involvement)
4.2 Ensuring Safe and Private Information in Caregiving		
- AI can help gather information in caregiving on how to perform tasks (e.g., catheter) or respond to specific cases (e.g., diabetes) - And manage care continuity and improve coordination across stakeholders (e.g., clients, caregivers, medical professionals, families)	- However, many were worried about AI making a mistake (e.g., sharing wrong information) in such a high-stakes (i.e., life-and-death) situation - Misleading AI outputs might also take advantage of vulnerable older adults - The need for more data could encroach on the privacy of sensitive health information	- Careful deployment mechanisms that ensure correctness/reliability/security of outputs (e.g., test run it first) - Creative combinations of machine outputs and human verification across stakeholder groups to repair/maintain trust
4.3 Pushing the Limits of Virtual and Physical Connections		
- AI could help bridge gaps of physical isolation inherent in home care across cooperatives, workers, and clients - And in some creative ways, AI could impact the physical environment (e.g., monitor clients when there aren't enough in-person caregivers, adjust climate controls, propose new arrangement of room for safety)	- There are limits to how much AI can do in the physical world - And concerns about whether AI could account for the unpredictability of the physical environment - There were also concerns around how this impacts the broader social (i.e., relationality) and physical (i.e., sustainability) environment	- Limiting AI to filling in for or facilitating physical human connections and actions, rather than replacing them - Making the physical/environmental costs more clear upfront for each action so people could better assess the cost-benefit ratio - Experimenting with different configurations that require less physical/environmental resources (e.g., smaller, more local models)
4.4 Raising Worker Voice and Challenging Structural Power		
- AI could help overcome some barriers of participating in the cooperative (e.g., social skills coaching, training on being a member owner, visual financial modeling/analysis) - And help pave the way for future cooperatives (e.g., home care launch box, interactive roadmap)	- But AI cannot overcome all of the barriers, such as the costs/pressures of having to balance concerns around care and business - And it collides with some of the entrenched power dynamics in home care as well as in society more broadly	- Cooperative structures as laboratories of innovation for governance given worker-centric models/worker-owner formation - Which expand from individual cooperatives to the cooperative movement and networks and finally to shape the future of work

Table 2. Summary of findings that describes some of the perceived benefits or harms of AI in certain facets of home care worker cooperatives paired with opportunities to further explore these tensions.

5 Discussion

5.1 Summary of Findings

Through design workshops with caregivers, administrators, and developers associated with a national network of home care worker cooperatives, this paper explores how different stakeholders currently experience AI and what they would like to see AI become in order to support their goals of care and cooperation. First, participants shared examples of how AI could help with improving the efficiency of tasks in these small businesses in the home care space where there are small margins and few but already overburdened caregivers. However, the participants also noted that AI systems would require additional resources, including human resources to ensure a caring touch. Second, participants identified ways AI could be helpful in gathering information but noted that the over-reliance on AI could be detrimental, given the high-stakes nature of healthcare and the sensitivity of data involved. Additionally, participants saw AI as a tool to help lessen the physical isolation of cooperatives, caregivers, and clients, but also cautioned about how the social and physical environment could impact and be impacted by AI systems. Finally, participants discussed how AI could help

build worker voice in the context of clinical care teams and workplaces. They also discussed how AI's effectiveness can be limited by existing power structures both in the workplace and more broadly in its design and governance. See Table 2 for a summary of the findings of some of the perceived benefits and harms of AI in the home care worker cooperative space and some ways these insights connect to the opportunities discussed further in the following sections.

5.2 Opportunities for Balancing Care and Cooperation

Much of the findings weave between the two considerations of care and cooperation, specifically in cases where there was a tension between care quality and work efficiency. We present a few key insights about how these two can be balanced in future instances of AI system development and governance.

5.2.1 Splitting Responsibilities Between Humans and AI. Participants mentioned that even though AI could be more efficient and more calm and collected than human workers, they were concerned about AI taking over all functions of care and making mistakes, as errors “*can be really dangerous when you look at caring for others*” (MA226). They especially discussed that, for any decision or output produced by AI systems, it is important to “*always, always, always have it verified by the humans*” (C5). This mirrors human-in-the-loop architecture that is discussed in literature, which prior research has shown is especially pertinent to “*high-stakes applications where full automation remains insufficient*” [45] and has proposed different models such as continual optimization [47] or data augmentation [98]. The participants in this study did reinforce these concerns for specifically the home care space, which is high-stakes (i.e., “*life and death*” (C6)) and also has high levels of uncertainty (e.g., scheduling in light of transportation issues and changing client requirements). However, they also brought up interesting perspectives that highlighted the gap between theoretical advancements in human verification and perceptions of whether the AI systems would be trustworthy or reliable, which may be based on public discourse about emerging generative AI systems as well as the participants’ own experiences of subpar systems. These impressions highlight important considerations for how systems developed for high-stakes, uncertain care contexts need to incorporate and communicate specific adaptations for safety to manage stakeholder expectations.

The participants also expressed worries about the limitations of AI systems in care contexts. They were skeptical about AI systems’ abilities to perform key functions of care, such as interacting with the physical world (e.g., the ability to do “*hands-on labor*” (C3)). Some also expressed concerns about how AI systems would not be able to maintain authentic relationships that are core to giving care, which would potentially result in changes to not only personal connections, but also the broader social fabric. Prior research has discussed how interpersonal elements are particularly difficult to automate, including referring to care work as the “*last bastion of ‘human’ work*” [53] that, in turn, becomes the “*last human job*” [67]. This echoes other research that emphasizes the difficulty of quantifying care work that is invisibilized [57, 82, 91]. The participants echoed these ideas on how caregiving is ultimately “*very human-centered work*” (C4). However, they also bring up another perspective where cooperatives present a different approach to this humanness, because as workers, they “*aren’t just a number for them*” (C5). This presents an interesting approach where having some element of cooperativity, such as through meaningful engagement of caregivers or clients, could bring more of a human element back into AI systems.

These concerns the participants expressed about the balance of humans and AI align with broader conversations about **the best configurations for human-AI collaboration, especially in cooperative, care-centered workplaces.** This is especially pertinent for organizations like home care worker cooperatives because of the low resources (i.e., being small businesses), high cost of experimentation (i.e., given concerns about safety), and barriers to change (i.e.,

being in an industry that has historically been hierarchical and slower to adopt technology). There is research about how the balance between which human-machine collaborations are “pro-worker” in terms of how they change labor productivity, value human expertise, or shift labor’s share of national income [2]. The participants of the study suggested different examples of “*labor-augmenting technologies*” and “*capital-augmenting technologies*” that would help these smaller businesses use their resources more efficiently and also suggested examples of “*expertise-leveling technologies*” where AI could be used for training, including for connection and communication with others. However, they did not specifically mention “*automation technologies*” given their concerns around AI safety or “*new task-creating technologies*” given the low appetite for new technological burden given how overburdened caregivers were already. These discussions scaffold future efforts to figure out more optimal combinations for human-AI collaboration in care and cooperation.

5.2.2 Reconciling Stakeholder Priorities and Perspectives. The perspectives different groups of stakeholders had about AI systems depended largely on their job functions and positionality. In broader strokes, more of the administrator and developer participants focused on potential functions of AI systems in improving the operational side of the business, whereas those coming from the caregiver perspective tended to focus on the role AI could have in direct care, reflecting a divide between organizational versus individual-level scopes. Additionally, those who had more internal (i.e., back office) focuses were more open to having AI tools take on more tasks whereas those who were more client-facing (i.e., front office) had more hesitations, due in part to the increased concern the participants had over AI systems’ limitations in physical, interpersonal care circumstances. For example, administrators and developers were more likely to speak more on the shortage of caregivers and how that leads to low margins and strapped resources, which created more space for AI systems to bring improvements and efficiencies. On the other hand, caregivers were more likely to speak more to their concerns about AI in terms of the safety of the clients.

However, in contrast to other settings where divides between perspectives about AI were more stark (i.e., more traditional worker-employer relationships), there were areas where both operational and caregiving stakeholders aligned. One example is that both sides were very concerned about the matching and scheduling process, even if they emphasized different aspects of it (i.e., operations emphasized caregiver shortage, caregivers underlined the impact it would have on care quality). In this case, an AI-powered system could help achieve these intertwined goals from both perspectives and the participants suggested systems that could improve communication so that “*everybody can be on the same page*” (C1). One key reason why there was greater alignment is because the nature of the cooperative requires caregivers to be to some degree involved in operations as worker-owners. In particular, the participants such as C2 who had experience on the board of their cooperatives were more acutely aware of the tension between needing to “*do what’s in the best interest of the business*” even if it does not make every caregiver or client happy.

Thus, home care cooperatives are uniquely situated because they are in an industry that has interconnected priorities (i.e., care quality and efficiency) and in organizations that are structured to bring together diverging voices and perspectives towards consensus. This creates an opportunity to **explore different approaches to the process of giving workers a say across the stages of the AI pipeline**, moving beyond having them serve only as “*rubber stamps*” (C6). This could include integrating the human-computer interaction focus on design stages [9] and with employee involvement as described by labor relations literature [48] and cooperative governance as explored in economics and social movements [50]. This would be important especially in ways that take into account the high burden these workers face from both the caregiving and administrative perspective, as described by prior attempts at designing data-driven technologies in the home care context [87]. Moreover, there are also opportunities to **explore what AI systems that span operations and caregiving could look like**, knowing that any technology that is meant to be used in the back

office would still impact caregivers not only in how they are able to provide care, but also how they can participate in and contribute to the cooperatives' operations.

5.2.3 Considering External Relationships and Impacts. The values of care and cooperation that the participants wanted to foster extended beyond themselves and their specific organizations. This manifested in the care they showed for their clients, including the care in which they described their relationships with their clients and in instances where the participants spent a lot of time brainstorming different applications that supported the clients. This reflects prior research that demonstrated how care workers often put their clients before themselves, even in cases relating to data privacy [61]. Moreover, this also re-emphasizes the idea in literature that care workers, given their disposition for the job will also tend to prioritize societal wellbeing [18]. Thus, their care also extended beyond the circle of stakeholders involved in the caregiving ecosystem. Many were involved deeply in their local area, voicing concerns about the impact AI systems could have on “*the communities we care about more broadly*” (D2) and specifically the land in which they were living (D1). They were especially excited about examples of community-driven development and community-owned infrastructure as ways to create more equitable and less wasteful AI systems.

Therefore, there is an exciting opportunity to explore **new community-driven paradigms that address concerns around environmental and other societal externalities**. This could include how these communities respond to AI systems, including through community-driven audits of AI outputs and impacts [90], or how communities handle AI infrastructure, including data (e.g., sovereignty [10], control [21]) and community ownership (e.g., Te Hiku Media [30]). Moreover, this could push forward alternative AI architectures that are more relevant for care and cooperation-focused organizations, such as small language models that have been used in human services [68] or carbon-aware computing [17]. Furthermore, this is an opportunity to explore **how to make short-term and long-term costs more prominent as care and cooperation-focused individuals and organizations consider their AI use**. This builds on long literature of how to change behavior around sustainability issues [8] and includes emerging research on how to integrate sustainability into engineering practices [24], simulations to demonstrate the potential impacts [31], and interactive calculators to help reason through impacts [13].

5.3 Limitations and Future Work

This study is the beginning of developing a greater understanding of how care and cooperation can be integrated into the design and governance of AI technologies. Moving forward, possible next steps include reaching broader samples of cooperatives and caregivers to explore the preliminary geographic trends we identified in this study (e.g., cooperatives in more remote areas) and the role of other organizational characteristics, such as size (e.g., smaller cooperatives that are just starting up). Future studies might also engage more diverse groups of participants to reflect the growing number of women and people of color in worker cooperatives and democratic workplaces [84] and the high proportion of women, immigrants, and people of color in the home care workforce [66]. Larger, more representative, samples would also enable the use of quantitative survey methods, especially by contributing to cross-sector longitudinal efforts to understand the adoption of AI in workplaces over time (e.g., Partnership on AI and Quality of work (PAIQ) [35]) and the perspectives and priorities of stakeholders at different stages of AI roll-out.

Another potential direction is exploring specific examples of technologies raised during the design workshop sessions, including areas in which AI may be useful. These include tools for reducing the overhead required for matching and scheduling caregivers and clients, communicating the value proposition of membership in a home care cooperative, training and educating new members about cooperative principles and care practices, facilitating conversations toward

caregiver consensus and client solidarity, and sharing and gathering information across cooperatives. Going more deeply into one of these ideas could bring together more people from a given cooperative for ideation, for example across functions of caregiver, administrator, and developer. This process could offer insights into what implementing some of the suggestions of what balancing human and AI efforts could look like. For example, this could concretize what exact functions and division of labor could look like between AI and humans in order to be the most efficient, building on broader questions that were raised by Pugh [67] and others interested in human-AI collaboration [4]. Another area is how conflicts between different stakeholder perspectives could be resolved during conversations or in proposed designs, which has been a concern in multi-stakeholder situations that center care [63], workers [32], and voice [41].

This research sets the groundwork for the design and governance of AI systems in similar work contexts that focus on care and the sharing of emotional labor/well-being (e.g., healthcare, counseling, education), cooperation and consensus/collectivity (e.g., other democratic workplaces, unions, nonprofits), or a combination of both; and relying on interpersonal relationships/power. This line of inquiry would contribute to emerging literature on how mission-driven and social good organizations balance the benefits and harms of AI systems, including how organizations have adopted AI and are currently responding to its promises and pitfalls (i.e., how it is currently adopted [72], how organizations have responded [42], guidelines for responsible practices [54]). Moreover, this research identifies future ways care and cooperation could be made part of AI systems [50].

6 Conclusion

This paper focuses on home care worker cooperatives as a unique site for exploring how worker voice could influence emerging technologies like AI. Through discussions and design activities, we identify key concerns and hopes that caregivers, administrators, and developers in the home care worker cooperative ecosystem have about their current experiences with AI and what they would like to see in the future. The participants discussed potentially using AI as a tool to make work more efficient, gather information effectively, communicate across borders, and shift power to caregivers. However, they also discussed some of the tensions in how AI should be used in conjunction to human checks and collaboration, with special attention paid to the social and environmental impacts of the technology. While this research does not necessarily resolve the tensions inherent in the design and governance of AI tools in the context of home care worker cooperatives, it does point to a path forwards with core questions that need to be resolved as well as present a few concrete design suggestions.

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